

Cycle-Jump Demonstration Using a Chaboche Unified Viscoplastic UMAT

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1.1 Motivation

Long cyclic simulations are computationally expensive when every loading cycle is resolved explicitly. A cycle-jump approach reduces the cost by identifying a stabilized per-cycle evolution of selected internal variables and extrapolating this evolution over many cycles. In the present demonstration, the cycle-jump variable is the accumulated viscoplastic strain stored by the UMAT as SDV1.

1.2 Abaqus-UMAT Model

The model was implemented in Abaqus/Standard using a single C3D8 block element smoke-test geometry. The material response was described by a Chaboche-v1 unified viscoplastic UMAT and loaded under displacement-controlled fully reversed cyclic loading. The state variable SDV1 represents the accumulated viscoplastic strain p , while SDV15 stores the last viscoplastic increment. This compact setup was selected to isolate the UMAT response, the Abaqus interface, and the cycle-jump postprocessing workflow.

1.3 Amplitude Selection

An amplitude sweep was first performed to identify a physically useful validation amplitude. The $\pm 0.1\%$ and $\pm 0.2\%$ cases remained elastic or near-elastic, whereas the $\pm 0.5\%$ case activated viscoplasticity at a moderate stress level. The $\pm 1.0\%$ case produced stronger plastic cycling. Therefore, the $\pm 0.5\%$ strain amplitude was selected for the cycle-jump demonstration.

1.4 Explicit 10-Cycle Baseline

The selected $\pm 0.5\%$ case was simulated explicitly for 10 cycles. The analysis completed with 507 increments, 0 cutbacks, 0 warnings, and 0 errors. The simulation produced nonzero stress, nonzero reaction force, and monotonic accumulated state-variable evolution. The selected hysteresis loops show that the loop shape becomes nearly repeatable after the first-cycle transient.

1.5 Stabilized Internal-Variable Increment

The total SDV1 value is cumulative because it represents accumulated viscoplastic strain. It should therefore increase monotonically and should not be used directly as a stabilization criterion. Instead, stabilization was assessed using the per-cycle increment ΔSDV1 . Cycles 2–10 were used as the stabilized reference window. Over this window, the mean increment was 0.007185465191, the standard deviation was $3.368213202 \times 10^{-6}$, and the relative range was 0.001429037192, corresponding to approximately 0.1429%.

Single-element cyclic UMAT validation model

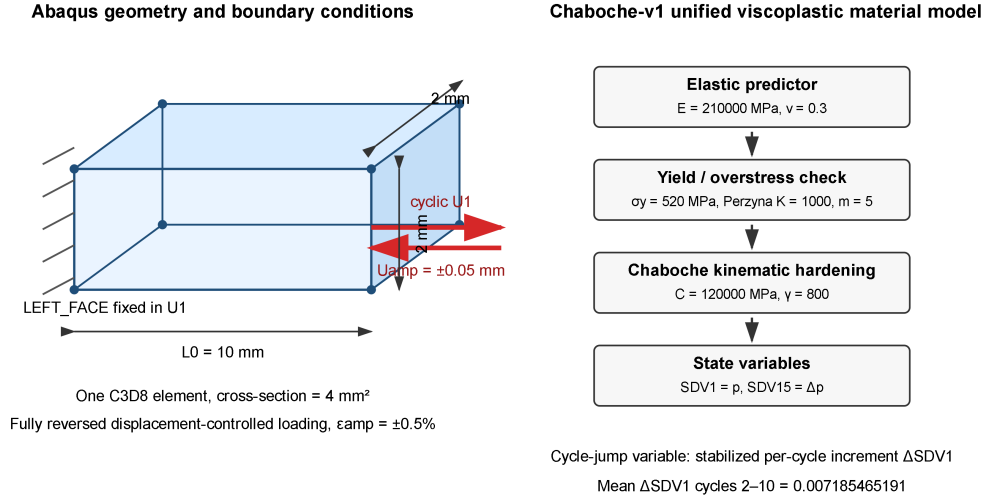


Figure 1: Single-element Abaqus validation model and Chaboche-v1 material-state workflow. The block is a one-element C3D8 specimen with $L_0 = 10 \text{ mm}$, a $2 \text{ mm} \times 2 \text{ mm}$ cross-section, fixed left face, and displacement-controlled cyclic loading on the right face.

Actual Abaqus C3D8 geometry extracted from input deck

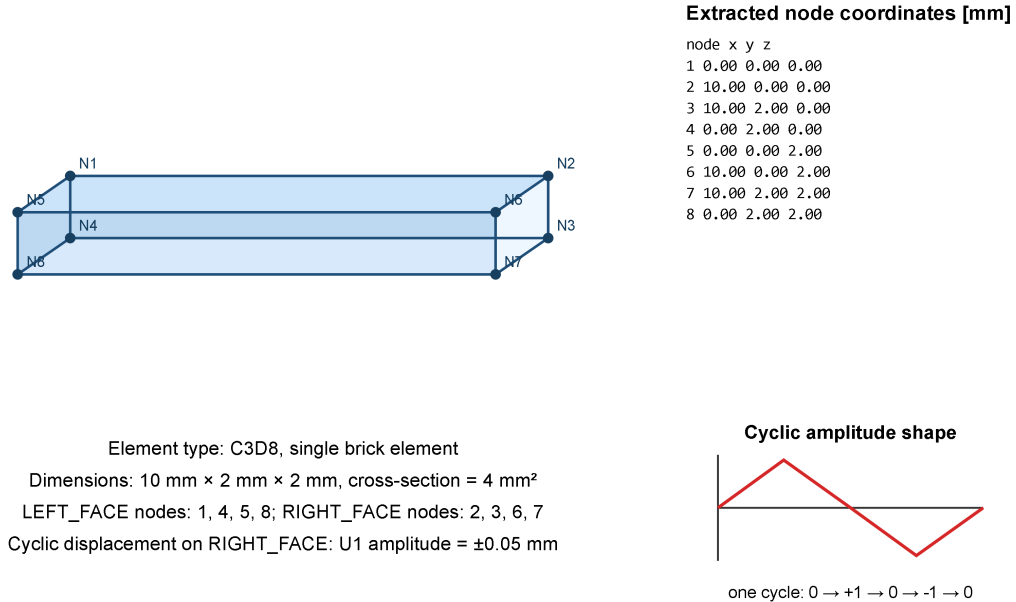


Figure 2: Actual Abaqus single-element geometry reconstructed from the input deck using the node coordinates and C3D8 element connectivity. The extracted model confirms the $10 \text{ mm} \times 2 \text{ mm} \times 2 \text{ mm}$ block, the left and right node sets, and the displacement-controlled cyclic loading direction.

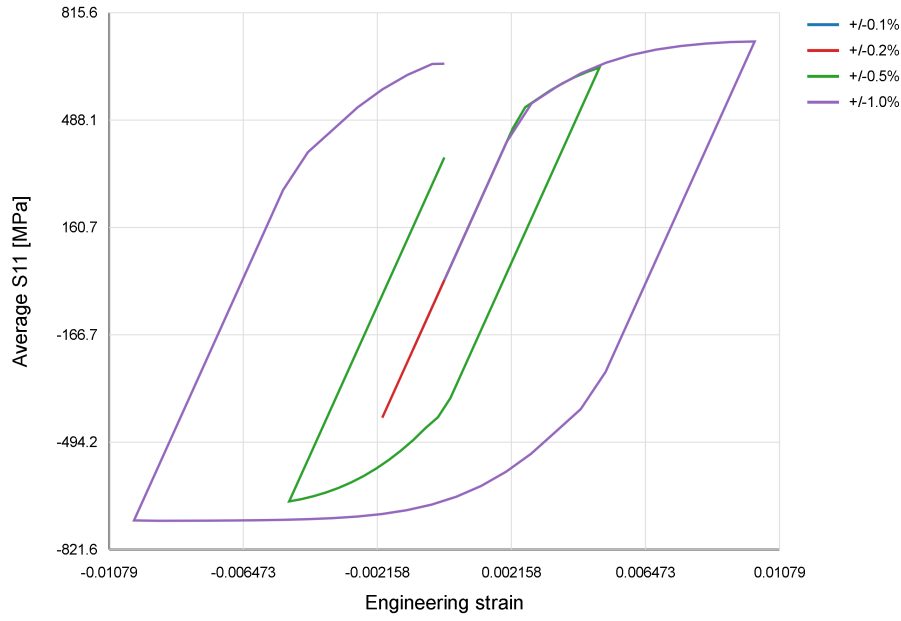


Figure 3: Amplitude sweep for the Chaboche-v1 UMAT. The smaller amplitudes remain mainly elastic, while the $\pm 0.5\%$ case activates viscoplasticity at a moderate stress level and was selected for the cycle-jump demonstration.

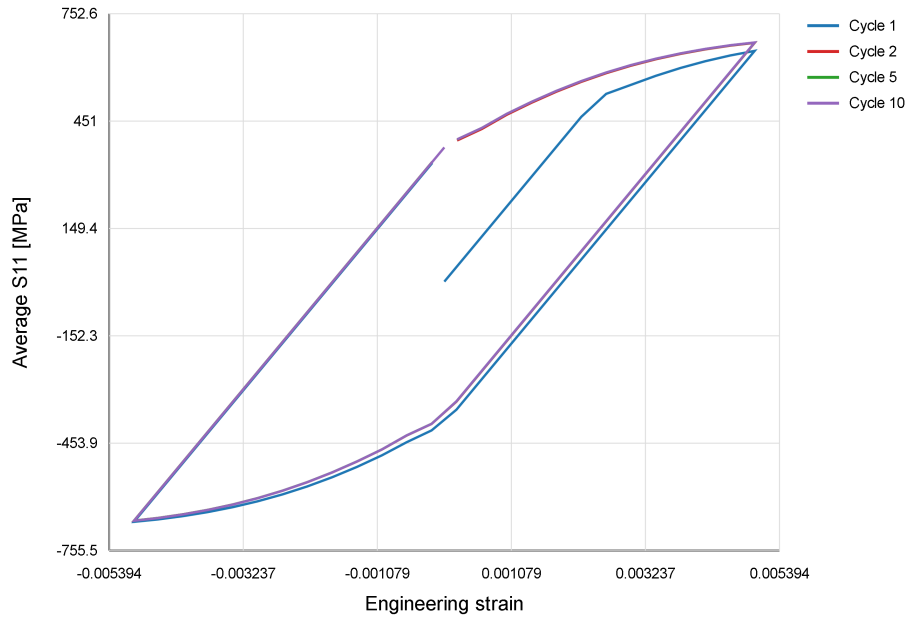


Figure 4: Selected hysteresis loops from the explicit 10-cycle Abaqus simulation at $\pm 0.5\%$ strain amplitude. The loop shape becomes nearly repeatable after the initial transient.

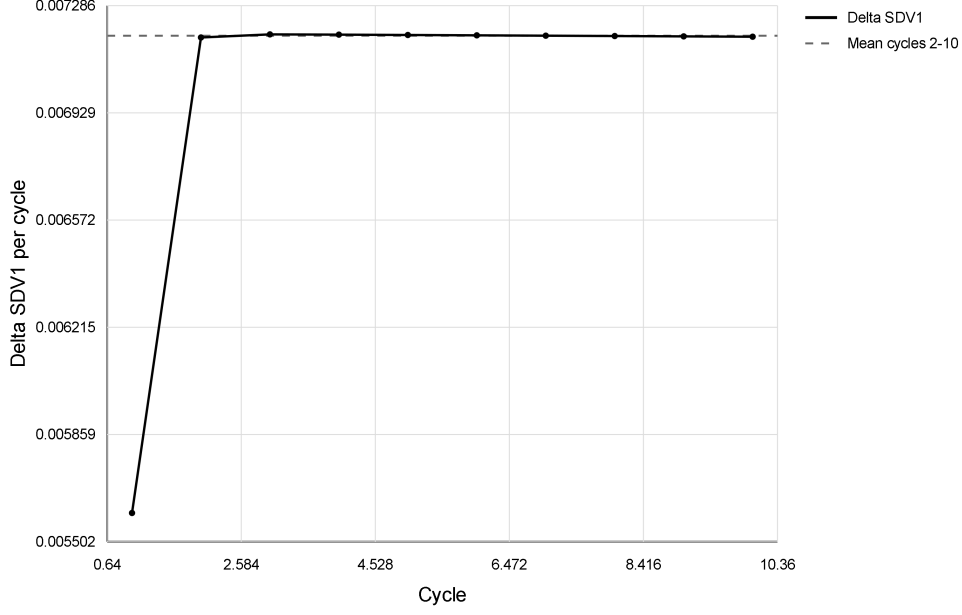


Figure 5: Per-cycle increment of accumulated viscoplastic strain. After the first cycle, the increment remains nearly constant, supporting the use of a cycle-jump predictor.

1.6 Cycle-Jump Predictor

The postprocessing-level cycle-jump predictor was defined as

$$p_N^{\text{pred}} = p_{10} + (N - 10) \overline{\Delta p_{2-10}},$$

where p_N^{pred} is the predicted accumulated viscoplastic strain at cycle N , p_{10} is the explicit value at cycle 10, and $\overline{\Delta p_{2-10}}$ is the mean per-cycle increment from cycles 2–10. The resulting predictions were $p_{20} = 0.1421214351$, $p_{50} = 0.3576853909$, $p_{100} = 0.7169586504$, $p_{200} = 1.435505169$, $p_{500} = 3.591144727$, and $p_{1000} = 7.183877322$.

1.7 Explicit 20-Cycle Validation

A separate explicit 20-cycle Abaqus simulation was performed to validate the cycle-jump prediction. This analysis completed with 1007 increments, 0 cutbacks, 0 warnings, and 0 errors. The predicted SDV1 value at cycle 20 was 0.1421214351, while the explicit Abaqus result was 0.1420256943. The absolute error was $-9.574084894 \times 10^{-5}$, corresponding to a relative error of 0.0674%.

1.8 Discussion and Limitations

This result validates a postprocessing-level cycle-jump predictor for the simplified Chaboche-v1 model and the selected $\pm 0.5\%$ strain-amplitude test case. The method does not yet inject jumped STATEV values into Abaqus and is not yet a calibrated fatigue-life model. The next implementation step would be state-variable injection using SDVINI, restart analysis, or another Abaqus workflow that initializes the material state after a cycle jump.

The Chaboche unified viscoplastic UMAT has therefore been successfully implemented and validated at the computational workflow level. It reproduces nonzero stress, reaction force, viscoplastic strain accumulation, cyclic hysteresis, and a stable per-cycle internal-variable increment suitable for cycle-jump prediction. However, the current Chaboche-v1 parameter set

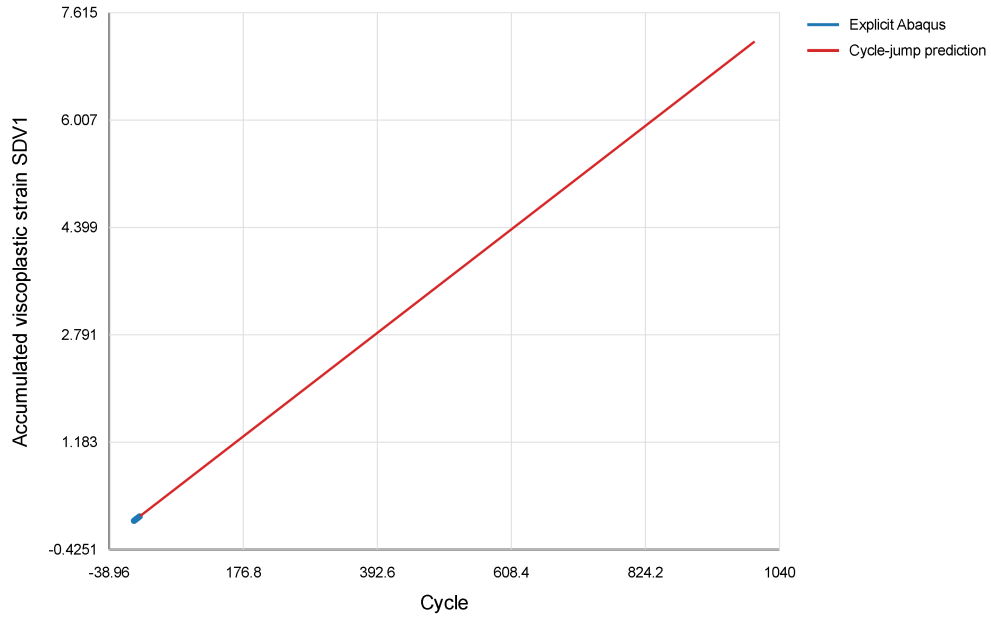


Figure 6: Cycle-jump prediction of accumulated viscoplastic strain using the stabilized per-cycle increment obtained from explicit cycles 2–10.

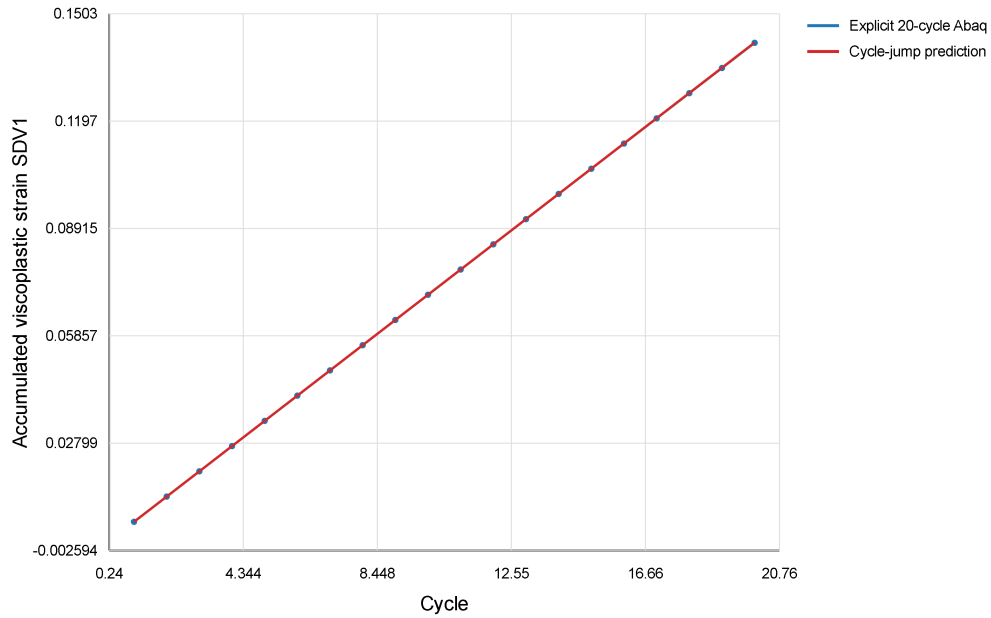


Figure 7: Validation of the cycle-jump predictor against an independent explicit 20-cycle Abaqus simulation. The predicted and explicit SDV1 values agree closely, with a relative error of 0.0674% at cycle 20.

should be treated as a demonstration and validation set, not yet as a fully calibrated 316 stainless steel material model.

1.9 Summary

The Chaboche-v1 UMAT, cyclic Abaqus workflow, postprocessing scripts, and cycle-jump predictor were validated. Explicit cycles 2–10 provided a stabilized per-cycle increment of accumulated viscoplastic strain, and the explicit 20-cycle check confirmed the prediction accuracy with only 0.0674% relative error in SDV1 at cycle 20.

2 Level-2 Preparation: Full-State Cycle-Jump Diagnostics

This section synthesizes the progression from Level-1 scalar SDV1 cycle-jump postprocessing to Level-2 preparation diagnostics for the simplified Chaboche-v1 unified viscoplastic UMAT. It explains the current findings and the rationale for deferring full STATEV injection to a later phase.

2.1 STATEV Inventory

The Chaboche-v1 UMAT manages 15 solution-dependent state variables (STATEV):

Index	Symbol	Physical meaning	Category
1	p	Accumulated viscoplastic strain	Active / required
2–4	X_{11}, X_{22}, X_{33}	Backstress normal components	Active / required
5–7	X_{12}, X_{13}, X_{23}	Backstress shear components	Near-zero
8–10	$E_{vp,11}, E_{vp,22}, E_{vp,33}$	Visc. strain normal	Active / required
11–13	$E_{vp,12}, E_{vp,13}, E_{vp,23}$	Visc. strain shear	Near-zero
14	RISO	Isotropic hardening stress	Recomputable
15	DP	Last visc. multiplier increment	Diagnostic

Table 1: STATEV inventory for Chaboche-v1 UMAT. STATEV(1–13) form the independent material state.

2.2 Full-State Cycle-History Extraction

Extract entire state vector STATEV(1–15) at each cycle end. Perform stability classification over cycles 2–10.

2.2.1 Key Findings

- STATEV(1): **stable extrapolation candidate**
- STATEV(2–4): **needs caution** (high cycle-to-cycle variation)
- STATEV(5–7): **near-zero**
- STATEV(8–10): **needs caution**
- STATEV(11–13): **near-zero**
- STATEV(14–15): **recomputable/diagnostic**

Full-state restart/injection cannot blindly extrapolate all components. The normal stress and strain components exhibit higher cycle-to-cycle variability than the scalar accumulated strain, indicating that phase-point consistency and accurate integration are critical for vector-valued cycle jumping.

2.3 Vector-Valued STATEV Cycle-Jump Control

Extend the scalar cycle-jump approach to a vector of active components: STATEV(1), STATEV(2–4), STATEV(5–7). Use first-order and second-order Taylor extrapolation with adaptive curvature checking to estimate a conservative global jump size.

2.3.1 Original Result

- Jump base: cycle 10
- Conservative global ΔN : **2**
- Adaptive target cycle: **12**
- Controlling component: **STATEV(2)** / X_{11}
- First-order SDV1 relative error at target: 0.0118%

The vector-valued analysis is more restrictive than scalar SDV1 jumping because the normal backstress components impose tighter constraints on the global jump.

2.4 Exact-Output Diagnostic Branch

The original extraction used nearest available ODB frames at integer cycle-end times. For phase-sensitive components (especially backstress and viscoplastic strain), this introduces time ambiguity. A separate “exact-output” Abaqus run was created with `TIME MARKS=YES` to force exact integer cycle-end frame output.

2.4.1 Exact-Output Results

- Maximum absolute cycle-end time error: 0 (exact frames) vs. original 0.00974
- Cycle-20 STATEV(1): 0.134750679135 (exact) vs. 0.142025694251 (original)
- Relative difference: **5.12%**

Key observation: The `TIME MARKS=YES` directive forced Abaqus to accept smaller time increments to align with requested output times. This altered the accepted increment sequence compared to the original run, causing the UMAT integration to produce a noticeably different cumulative viscoplastic strain by cycle 20.

2.4.2 Exact Vector Result

- Conservative global ΔN : **1** (vs. original 2)
- Adaptive target cycle: **11** (vs. original 12)
- Controlling component: **STATEV(2)** / X_{11} (unchanged)

2.4.3 Interpretation

The exact-output run successfully demonstrated that phase-point ambiguity can be removed by requesting exact output marks. However, the 5.12% difference in cycle-20 SDV1 indicates that the simplified Chaboche-v1 UMAT is **increment-schedule sensitive**. The exact-output run cannot replace the original validation baseline because it represents a different computational path (forced smaller increments) with a measurably different material response. Therefore, the original 20-cycle result remains the main validation reference, while the exact-output branch is used to demonstrate phase sensitivity and the need for caution in vector-valued STATEV cycle jumping.

2.5 Why STATEV Injection is Deferred

2.5.1 Status

- Level 1 (postprocessing SDV1 prediction): validated with 0.0674% error; thesis-ready as demonstration case
- Level 2a (full-state extraction): completed; inventory and stability classification available
- Level 2b (vector-valued control): completed; shows vector approach is more conservative than scalar
- Level 2c (exact-phase diagnostics): completed; reveals increment-schedule sensitivity

2.5.2 Robustness Requirement

1. **Increment-schedule sensitivity:** The exact-output comparison revealed that the Chaboche-v1 UMAT response is sensitive to the number and size of time increments. An injected state at a given cycle could produce different future predictions depending on the integration scheme used in the next Abaqus run.
2. **Vector-valued complexity:** The vector cycle-jump approach requires consistent treatment of 13 independent components. Without first reducing the UMAT's increment sensitivity, injecting a state vector carries high risk of uncontrolled accuracy loss.
3. **Thesis narrative alignment:** The demonstrated Level-2 preparation diagnostics are themselves strong thesis content. Stopping here allows a clear discussion of why material model integration robustness is necessary before advanced continuation methods.

2.6 Distinction Between Method Levels

Level 1 (Postprocessing): Extract computed STATEV history from existing ODB. Fit polynomial or first-order model to cycle-end values. Extrapolate and predict cycle-end state at skipped cycles. Read-only ODB access; no material model changes. Postprocessing-level accuracy.

Level 2 (Preparation): Extract full state vector. Perform stability and sensitivity analysis. Diagnose increment-schedule dependence. Design injected-state continuation logic. Diagnostic runs only; no material model changes. Diagnostic-level accuracy.

Level 3 (Full Integration): Implement injected-state restart in Abaqus. Modify UMAT or use Abaqus restart keywords. Run accelerated multi-cycle segments with injected STATEV. Active Abaqus workflows and state-variable management. Subject to robustness of UMAT and increment selection.

3 Increment-Schedule Sensitivity of the Chaboche-v1 UMAT

This section freezes the Stage 3 sensitivity study that compared three controlled DMAX settings for the same 20-cycle Chaboche-v1 simulation.

The question is whether accumulated viscoplastic strain remains invariant under time-increment refinement. It does not. The cycle-20 accumulated strain increases monotonically as DMAX decreases, so the UMAT remains increment-size sensitive and Level-3 STATEV injection stays deferred.

Case	DMAX	INC limit	STATEV1 @ cycle 20	Rel. diff. vs ref.	Avg S_{11} @ cycle
Original-output baseline	0.020	4000	0.142025694251	0%	376.434143
DMAX-refined case	0.010	4000	0.143569096923	1.0867%	380.566436
Fine DMAX case	0.005	6000	0.145257070661	2.2752%	385.053924

Table 2: Stage 3 increment-schedule sensitivity summary for the Chaboche-v1 UMAT. The reference value for STATEV1 at cycle 20 is 0.142025694251.

3.1 Cycle-20 Summary

3.2 STATEV1 Versus DMAX

The first sensitivity plot shows the final cycle-20 accumulated viscoplastic strain against the maximum allowed increment size. The trend is monotonic: smaller DMAX produces larger accumulated STATEV1.

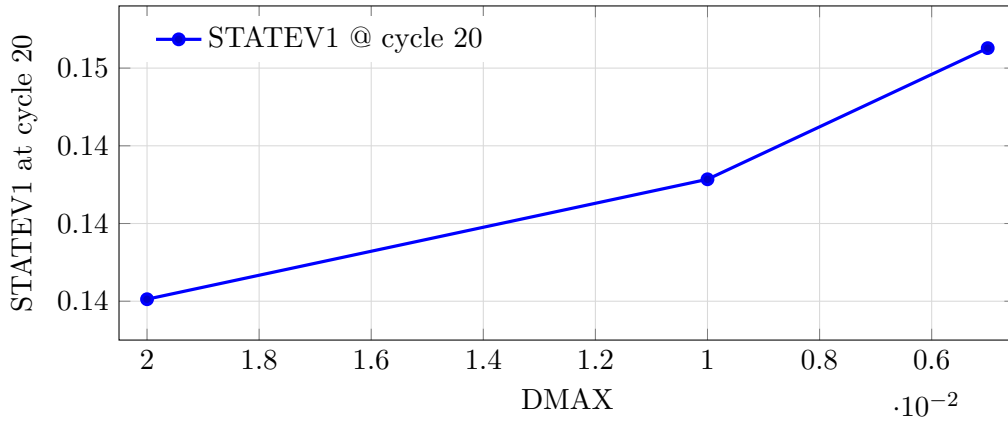


Figure 8: Cycle-20 STATEV1 versus DMAX for the three completed Stage 3 runs.

3.3 STATEV1 Evolution Across Cycles

The second plot shows the cycle-end evolution of STATEV1 for the three completed cases. It confirms that the cases separate progressively as the increment schedule becomes finer. The source SVG versions of the Stage 3 plots are archived in the thesis package figures folder alongside this LaTeX-generated version of the same result.

3.4 Interpretation

The Chaboche-v1 UMAT shows a monotonic increase in accumulated viscoplastic strain as DMAX decreases, confirming increment-size sensitivity. The DMAX = 0.005 run required a larger increment limit because the earlier INC = 2500 deck failed with too many increments; the corrected INC = 6000 run completed successfully and became the final Stage 3 data point. The practical implication is unchanged: Level-3 STATEV injection and a full Nesnas–Saouni restart-level cycle jump remain deferred until UMAT integration robustness is improved.

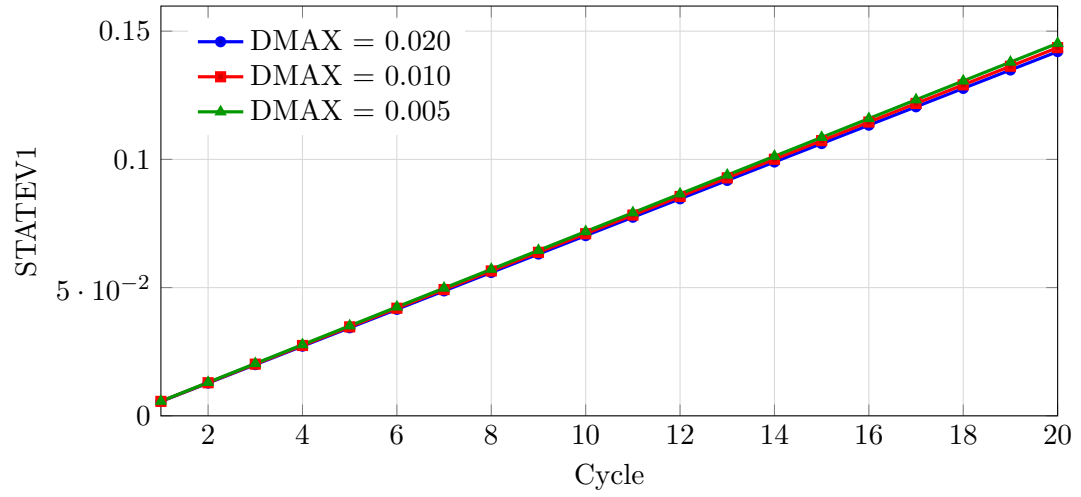


Figure 9: Cycle-end STATEV1 evolution for the three completed increment-schedule cases.